SIMATS ENGINEERING ASSESSMENT TOOL 2

COURSE NAME: INTERNET PROGRAMMING **Shanmukh.B**
192372217

COURSE CODE: CSA4303

Dr.Jaya mabel rani.A

COURSE OUTCOME COVERED

CO-1: Develop well-structured, standards-compliant web pages using HTML/XHTML and XML, and apply Internet protocols and HTTP communication mechanisms for web content delivery. **(L2)**

Assessment Tool: Data Interpretation (Medium)

Weightage: 20%

QUESTIONS: **Total Marks: 20**

1: HTTP Response Analysis

A web server returns the following HTTP response header:

```
HTTP/1.1 200 OK
Date: Mon, 01 Apr 2026 10:00:00 GMT
Content-Type: text/html
Content-Length: -150
Connection: keep-alive
```

Questions:

1. Identify incorrect or invalid fields: The Content-Length: -150 field is invalid because content length cannot be negative. The other fields are generally valid.

2. Analyze Content-Length: A negative content length violates HTTP requirements and may cause the response to be rejected or processed incorrectly. It should contain the actual non-negative body size in bytes.

3. Does the status code match? 200 OK indicates that the request succeeded, but it does not guarantee that every header is valid. The response headers must still be corrected.

4. Suggest corrections: Replace Content-Length: -150 with the actual body length, such as Content-Length: 150 if the body is 150 bytes. Keep the status code if the request succeeded.

2: **DOM Structure Analysis**

Consider the following DOM tree representation:

```
<html>
  <head>
  <title>TestPage</title>
</head>
  <body>
  <div>
  <p>HelloWorld
  </div>
 </body>
 </html>
```

Questions:

1. Identify missing or improperly closed elements: The element is missing its closing tag. The and other elements should be properly nested and closed.

2. How will the browser interpret it? Browsers attempt to automatically repair malformed HTML and construct a DOM tree. However, the resulting structure may differ from the intended structure.

3. Impact on rendering and layout: Improper nesting may affect spacing, element placement, and styling. It can also cause unexpected behavior when scripts access the DOM.

4. Suggest corrections: Add before and ensure all elements are correctly nested. Use the corrected HTML code shown above.

3: GET vs POST in Login System

A login system uses two different methods:

Method	URL Example	Data Visibility	Security Level
GET	/login?user=admin&pass=1234	Visible in URL	Low
POST	/login	Hidden in body	Moderate

Questions:

1. Differences between GET and POST: GET sends data through URL query parameters, making it visible in the address bar. POST sends data in the request body, so it is not normally shown in the URL.

2. Which method is suitable for login? POST is more suitable because credentials are sent in the request body rather than URL parameters. HTTPS is still necessary to protect credentials during transmission.

3. Security risks of GET: Credentials in URLs may be stored in browser history, server logs, and shared links. This can expose sensitive login information.

4. Best practices for secure login: Use POST over HTTPS, hash passwords securely on the server, and validate user input. Use secure session cookies and appropriate authentication protections.

4: HTML Code Inspection

Analyze the following HTML snippet:

```
<!DOCTYPE html>
<html>
<head>
<title>Sample</title>
</head>
<body>
<h1>Welcome
<p>This is a paragraph
<img src="image.jpg">
</body>
</html>
```

Questions:

1. Identify missing closing tags: The <h1> and elements are missing their closing tags. They should be closed using </h1> and respectively.

2. Analyze the <img> tag: The image tag is missing an alt attribute, which provides alternative text if the image cannot load. The <img> element is a void element and does not require a closing tag.

3. How will browsers handle these errors? Browsers generally attempt to repair missing closing tags and display the page. However, the page may not render exactly as intended.

4. Suggest corrected code: Add the missing closing tags and provide meaningful alternative text in the alt attribute. The corrected HTML is shown above.

5: Browser-Server Communication Flow

The following sequence describes a web request:

1. User enters a URL in the browser
2. DNS lookup occurs
3. HTTP request is sent
4. Server processes request
5. Response is returned
6. Browser renders page

Questions:

1. Interpret each step: The browser resolves the domain using DNS, connects to the server, and sends an HTTP request. The server processes it, returns a response, and the browser renders the page.
2. Where can delays occur? Delays can occur during DNS lookup, network connection, server processing, and data transfer. Large images, scripts, and stylesheets can also slow page rendering.
3. Role of DNS: DNS translates a human-readable domain name into an IP address used to locate the server. Without a successful resolution, the browser may not be able to reach the website by its domain name.
4. Suggest performance optimizations: Use DNS caching, optimized images, browser caching, and a content delivery network (CDN). Reduce unnecessary HTTP requests and optimize server response time.

Code of Conduct:

I _____Shanmukh.B(192372217)_____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Shanmukh.B

RUBRICS FOR CALCULATION:

Criteria	Weigh tage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Data	25%	Accurately interprets all data patterns, trends, and relationships	Interprets data correctly with minor gaps	Partial understanding; misses key trends	Misinterprets or fails to understand data
Analysis	25%	Provides deep analysis with strong logical reasoning and justification	Provides reasonable analysis with some justification	Basic analysis with weak reasoning	No meaningful analysis provided
Application of Concepts	20%	Effectively applies relevant concepts/tools to interpret data accurately	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts to data
Conclusion	20%	Draws clear, meaningful conclusions with strong insights and implications	Provides valid conclusions with moderate insights	Conclusions are vague or weak	No clear or valid conclusions
Clarity	10%	Presents interpretation clearly using proper structure, visuals, and terminology	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand

1: HTTP Response Analysis

```
<!DOCTYPE html>
<html>
<head>
   <title>HTTP Response Analysis</title>
</head>
<body>
   <h1>HTTP Response Analysis</h1>
   <p>Status: 200 OK</p>
   <p>Content-Type: text/html</p>
   <p>Content-Length: 150</p>
   <p>Connection: keep-alive</p>
</body>
</html>
```

Output :

HTTP Response Analysis

Status: 200 OK

Content-Type: text/html

Content-Length: 150

Connection: keep-alive

2: DOM Structure Analysis

```
<!DOCTYPE html>
<html>
<head>
   <title>TestPage</title>
</head>
<body>
   <div>
      <p>Hello World</p>
   </div>
</body>
</html>
```

Output :

Hello World

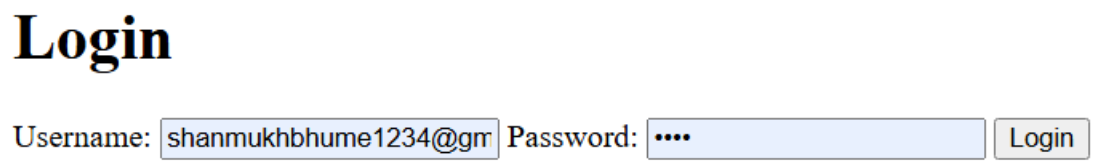
3: GET vs POST in Login System

```
<!DOCTYPE html>
<html>
<head>
   <title>Login System</title>
</head>
<body>
   <h1>Login</h1>
   <form action="/login" method="post">
      <label for="user">Username:</label>
      <input type="text" id="user" name="user" required>

      <label for="pass">Password:</label>
      <input type="password" id="pass" name="pass" required>

      <button type="submit">Login</button>
   </form>
</body>
</html>
```

Output :



4: HTML Code Inspection

```
<!DOCTYPE html>
<html>
<head>
   <title>Sample</title>
</head>
```

```
<body>
    <h1>Welcome</h1>
    <p>This is a paragraph.</p>
    <img src="image.jpg" alt="Sample image">
</body>
</html>
```

Output :

Welcome

This is a paragraph.

Sample image

5: Browser-Server Communication Flow

```
<!DOCTYPE html>
<html>
<head>
    <title>Browser-Server Communication</title>
</head>
<body>
    <h1>Browser-Server Communication Flow</h1>
    <ol>
        <li>User enters a URL in the browser.</li>
        <li>DNS resolves the domain name to an IP address.</li>
        <li>Browser sends an HTTP request.</li>
        <li>Server processes the request.</li>
        <li>Server returns an HTTP response.</li>
        <li>Browser renders the web page.</li>
    </ol>
</body>
</html>
```

Output :

Browser-Server Communication Flow

1. User enters a URL in the browser.
2. DNS resolves the domain name to an IP address.
3. Browser sends an HTTP request.
4. Server processes the request.
5. Server returns an HTTP response.
6. Browser renders the web page.